reconciliation.

Table 4.1 summarizes key skills for implementing declarative UIs, from core principles to advanced optimizations.

Table 4.1 Implementing declarative UI in React

Topic	Fundamentals	Applied skills	Advanced
Component-based architecture	Understand modularity and reusability principles.	Decompose UI into reusable components.	Architect large-scale apps with design systems.
JSX	Write HTML-like syntax in JavaScript/TypeScript. See appendix A.3 for best practices.	Use dynamic expressions, loops, and conditions.	Optimize JSX with fragments, portals, and advanced patterns.
Virtual DOM & Reconciliation	Explain how VDOM optimizes updates.	Trace how diffing/reconciliation work during renders.	Debug performance bottlenecks caused by reconciliation
Fiber architecture & React compiler	Understand Fiber's concurrent rendering model.	Apply concurrent features like transitions and Suspense.	Optimize rendering pipelines and React Compiler integrations.

Having covered the basics of declarative UI, let's dive into interview questions.

Initial question (fundamental skill): "Can you explain what a declarative